Highlights

A signal-to-drift ratio for quantifying winding-fault detectability and selecting condition monitoring across three wind-turbine generator types

Hüseyin Tayyer Canseven

- A scale-free signal-to-drift ratio measures fault detectability without a detector
- Identical winding faults are three different physical events in PMSG, SCIG and WFSG
- The fault signature relocates with the machine–drive combination, not the controller
- Voltage transducers lower the PMSG's detectable extent and add nothing for the SCIG
- Detectors transfer across generator types only with trial-referenced features